Customer Behavior Analysis

Executive Summary Report

Dataset	Analysis	Tools
250,000 Records · 13 Columns	EDA + RFM Segmentation	Python · Pandas · Seaborn · Matplotlib

Project Links

GitHub Repository: amegha-vs/Customer-Behavior-Analysis

Dataset (Google Sheets): ecommerce_customer_data_large

Introduction

Understanding customer purchasing behavior helps businesses improve marketing strategies, increase sales, and enhance customer satisfaction.

This project analyzes customer demographic and purchasing data to identify patterns in spending behavior. The analysis focuses on exploring relationships between customer age, income, loyalty status, and purchase amounts.

The goal is to extract meaningful insights that can help businesses make better decisions regarding customer targeting and retention.

Key Insights

1. **Product Category Performance**

   The analysis shows that certain product categories generate significantly higher revenue compared to others. This indicates that customer demand is concentrated in specific product segments, which play a major role in overall sales performance.

2. **High-Value Customers Drive Revenue**

   The top customer analysis reveals that a small group of customers contributes a large portion of total revenue. This highlights the importance of high-value customers and their impact on business profitability.

3. **Customer Segmentation through RFM Analysis**

   RFM segmentation identifies distinct customer groups based on their purchasing behavior. Some customers demonstrate high frequency and spending levels, indicating loyal customers, while others show lower engagement and may be at risk of churn.

4. **Customer Demographics and Spending Behavior**

   The age distribution and age vs purchase analysis indicate that customers from a wide range of age groups interact with the platform, with no strong linear relationship between age and spending amount. This suggests that purchasing behavior is influenced more by product demand than demographic factors.

5. **Payment Preferences**

   The payment method distribution highlights clear customer preferences for certain payment options. Understanding these preferences can help businesses ensure smooth and efficient payment processing for customers.

6. **Revenue Variability Across Product Categories**

   The revenue distribution analysis shows that some product categories have higher variability in purchase amounts, indicating occasional high-value purchases and diverse customer spending patterns.

7. **Customer Churn Patterns**

   The churn analysis reveals that a portion of customers eventually stop engaging with the business. Monitoring churn trends is essential for maintaining customer retention and long-term revenue stability.

8. **Sales Trends Over Time**

   The monthly revenue trend analysis highlights fluctuations in sales across different months and years. These patterns may reflect seasonal demand, promotional campaigns, or changing customer behavior over time.

9. **Feature Relationships**

   Correlation analysis indicates that total purchase amount is strongly influenced by product price and quantity purchased, while other factors such as age show weaker relationships with revenue.

Business Recommendations

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1. **Focus on High-Revenue Product Categories**

   Businesses should prioritize marketing and inventory management for product categories that consistently generate higher revenue, ensuring product availability and targeted promotions.

2. **Strengthen Relationships with High-Value Customers**

   Since a small group of customers contributes significantly to total revenue, companies should implement loyalty programs, personalized offers, and exclusive benefits to retain these valuable customers.

3. **Implement Targeted Marketing Based on Customer Segments**

   Using RFM segmentation, businesses can design customized marketing strategies for different customer groups, such as rewarding loyal customers and re-engaging customers who show declining purchase frequency.

4. **Improve Customer Retention Strategies**

   To reduce customer churn, businesses should focus on improving customer engagement through personalized communication, promotional offers, and proactive customer support.

5. **Leverage Sales Trend Insights for Strategic Planning**

   Monthly revenue trends should be used to plan marketing campaigns, promotional events, and inventory management strategies to maximize sales during peak periods and boost engagement during slower months.

Key Visualizations

Outlier Detection (Boxplots)

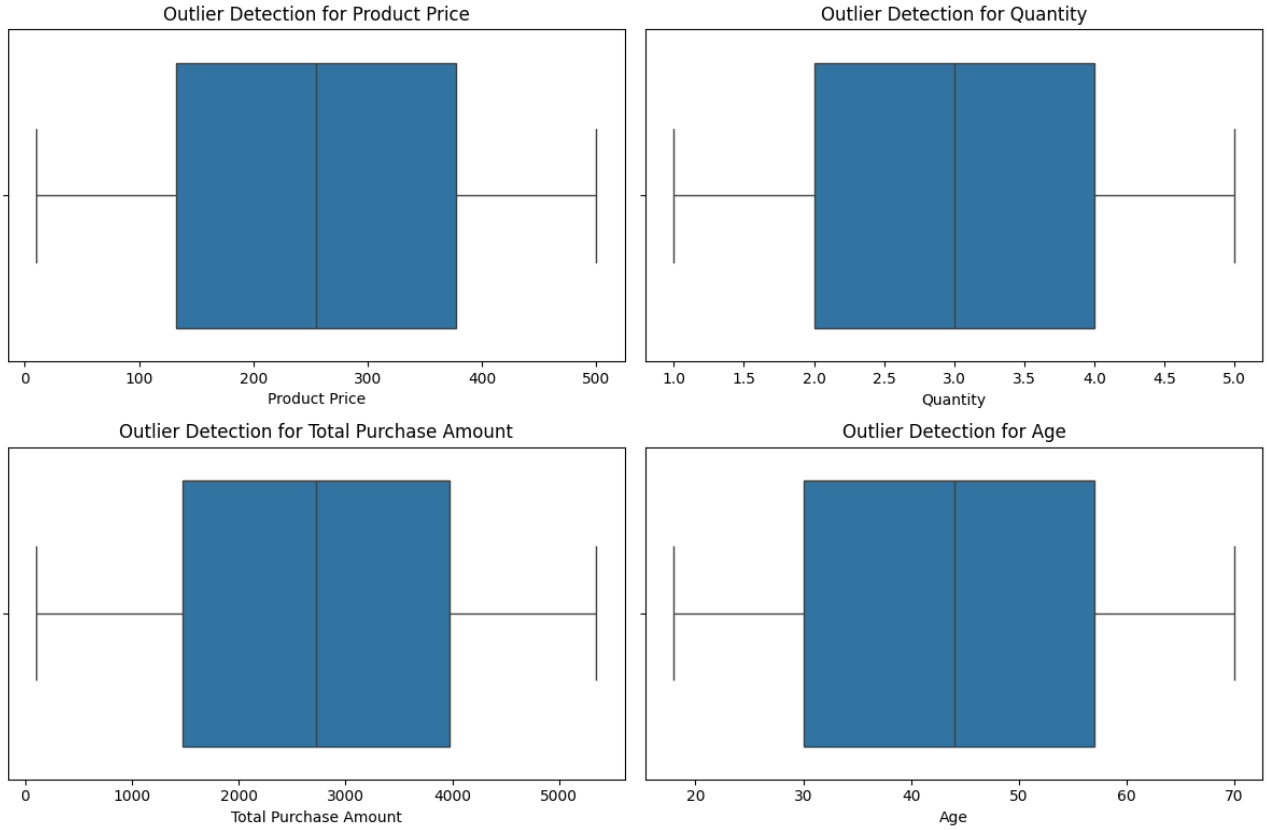

Fig 1: No outliers detected across Product Price, Quantity, Total Purchase Amount, and Age.

Product Category Distribution & Payment Method Usage

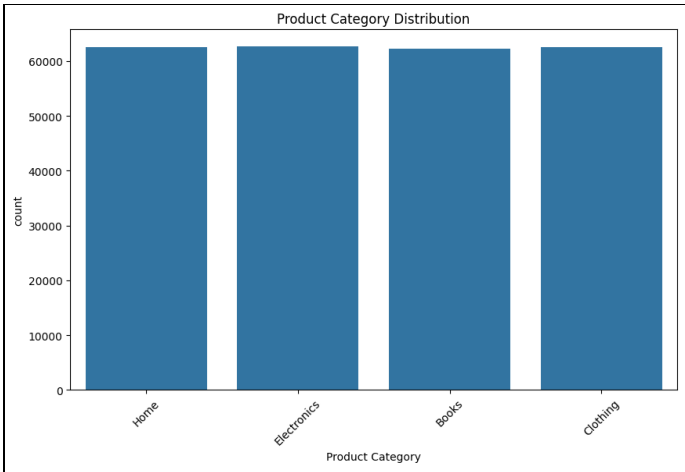

Fig 2: Product Category Distribution

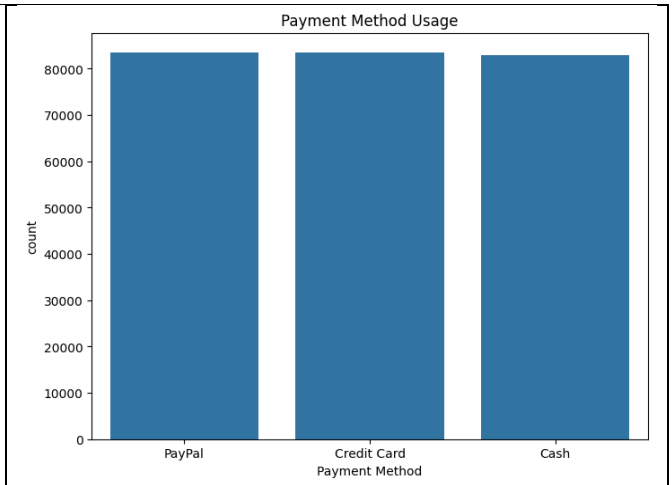

Fig 3: Payment Method Usage

Customer Age Distribution & Revenue by Product Category

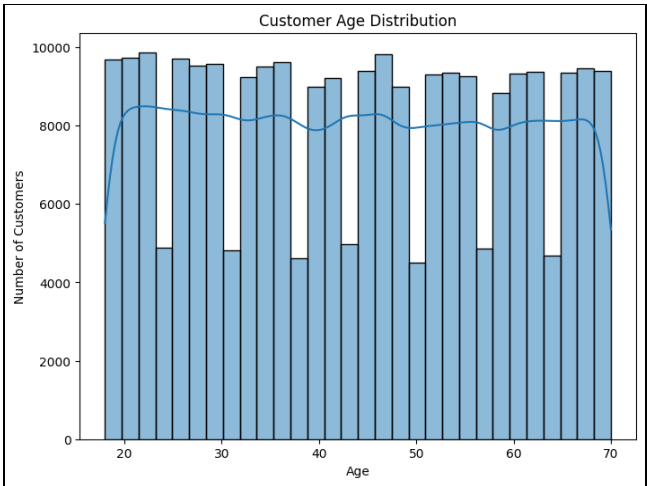

Fig 4: Customer Age Distribution

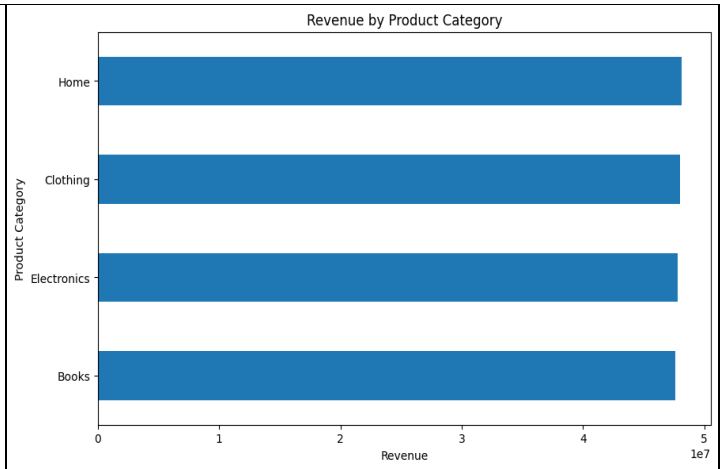

Fig 5: Revenue by Product Category

Correlation Matrix of Numerical Features

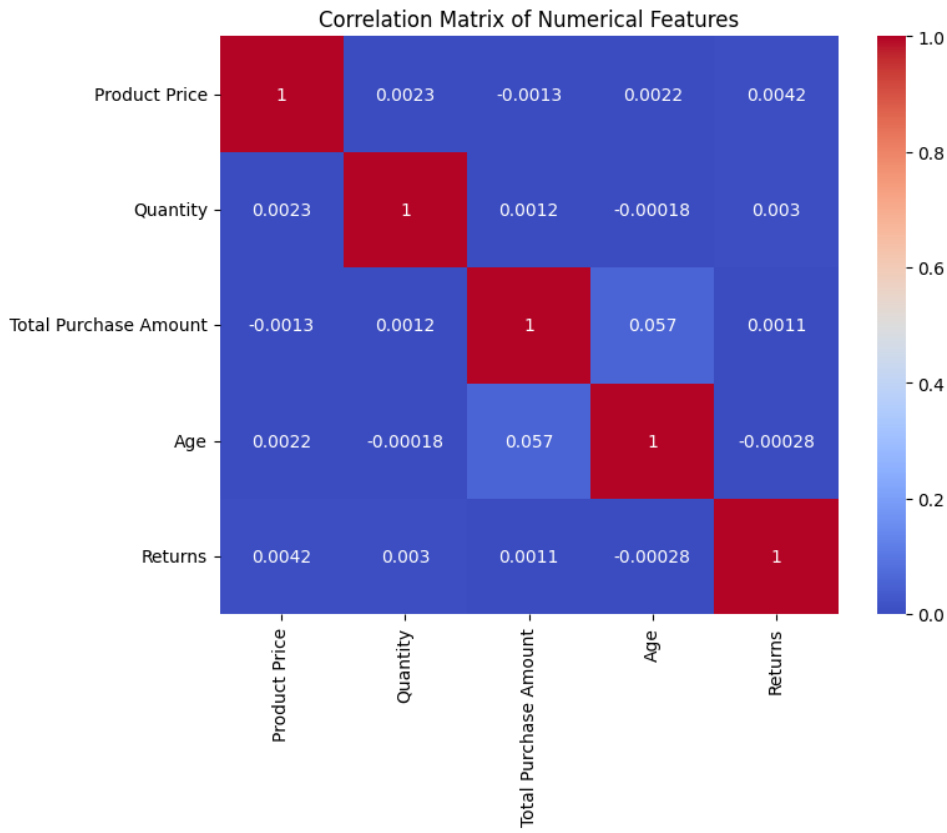

Fig 6: Total Purchase Amount strongly correlates with Product Price and Quantity.

Top 10 Customers by Revenue & Revenue Distribution by Category

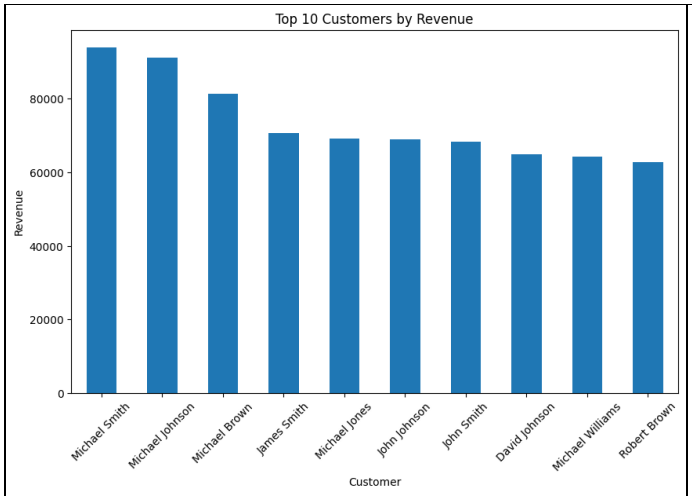

Fig 7: Top 10 Customers by Revenue

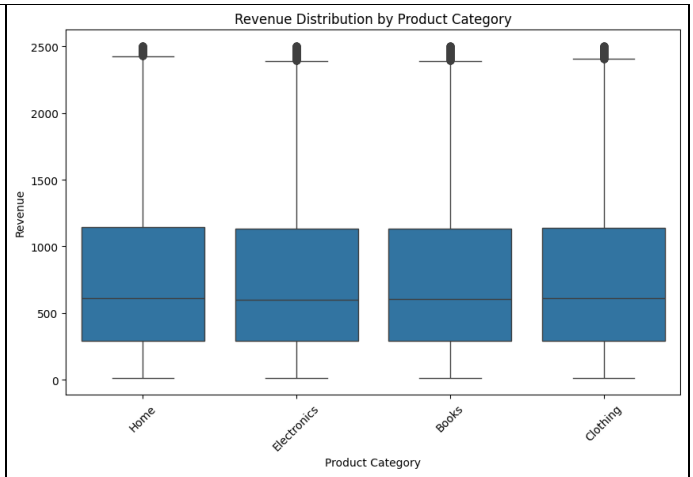

Fig 8: Revenue Distribution (Boxplots)

Customer Age vs Purchase Amount & Customer RFM Score Distribution

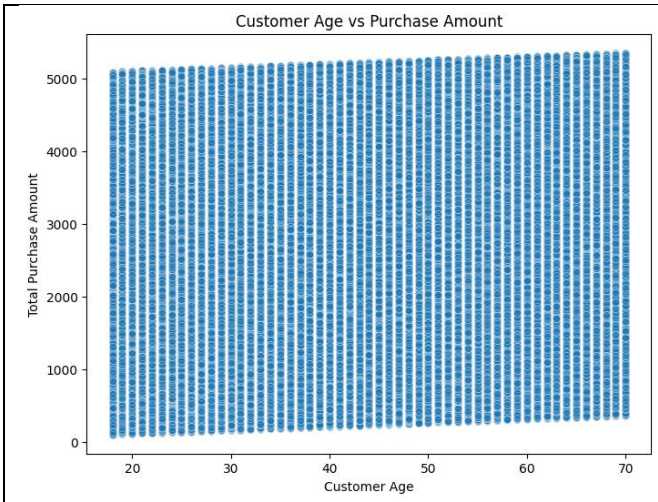

Fig 9: Customer Age vs Purchase Amount

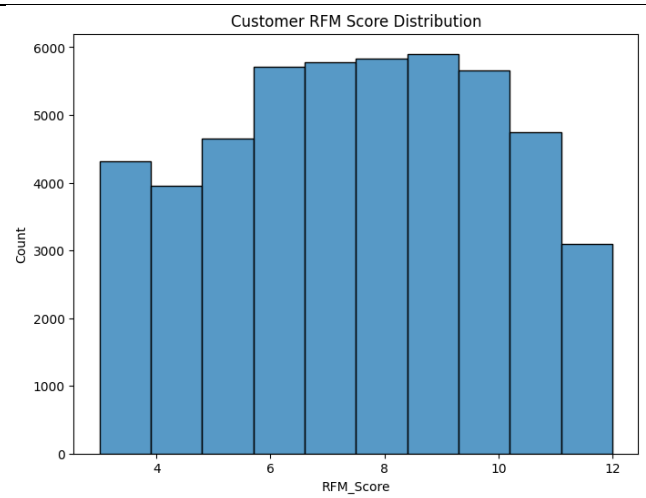

Fig 10: RFM Score Distribution

Customer Churn Distribution & Monthly Revenue Trends

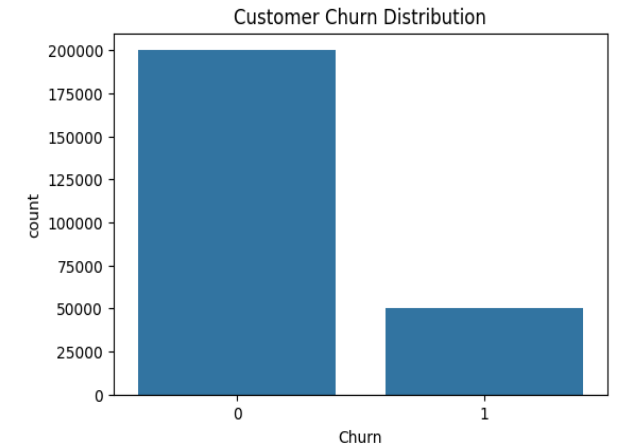

Fig 11: Customer Churn Distribution

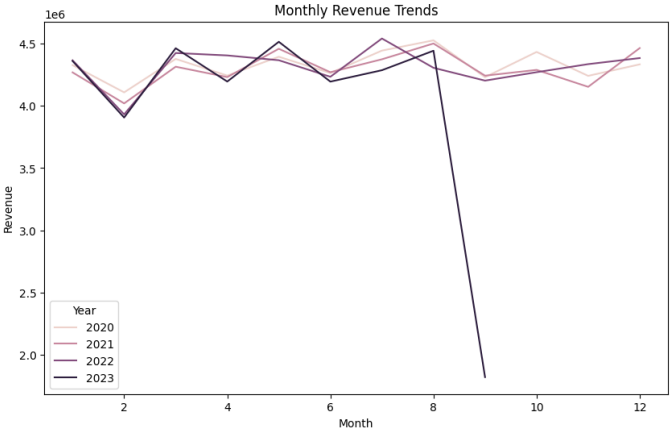

Fig 12: Monthly Revenue Trends